

Lami's Theorem and Concurrent Force Systems in Static Equilibrium

High-Level Overview

In **Engineering Mechanics**, analyzing static equilibrium involves understanding how forces interact to keep a body completely at rest. When multiple forces act on an object such that their lines of action all intersect at a single common point, they are called **concurrent forces**. If all these forces lie within the same two-dimensional plane, they form a **coplanar concurrent force system**.

Lami's Theorem is a powerful mathematical principle derived from the law of sines. It offers a fast, simplified approach to solving coplanar, concurrent three-force equilibrium problems without needing to resolve forces into orthogonal x and y rectangular components.

Fundamental Concepts & Core Terminology

Before applying Lami's Theorem, it is essential to understand the foundational principles governing force systems in statics:

- Force ($\vec{F}$):** A vector quantity representing an interaction that changes or tends to change the state of rest or motion of a body. Measured in Newtons (N) or kiloNewtons (kN).
- Mass vs. Weight:** Mass (m) is a measure of the amount of matter in an object, expressed in kilograms (kg). Weight (W) is the gravitational force exerted on that mass:

$$W = m \cdot g$$

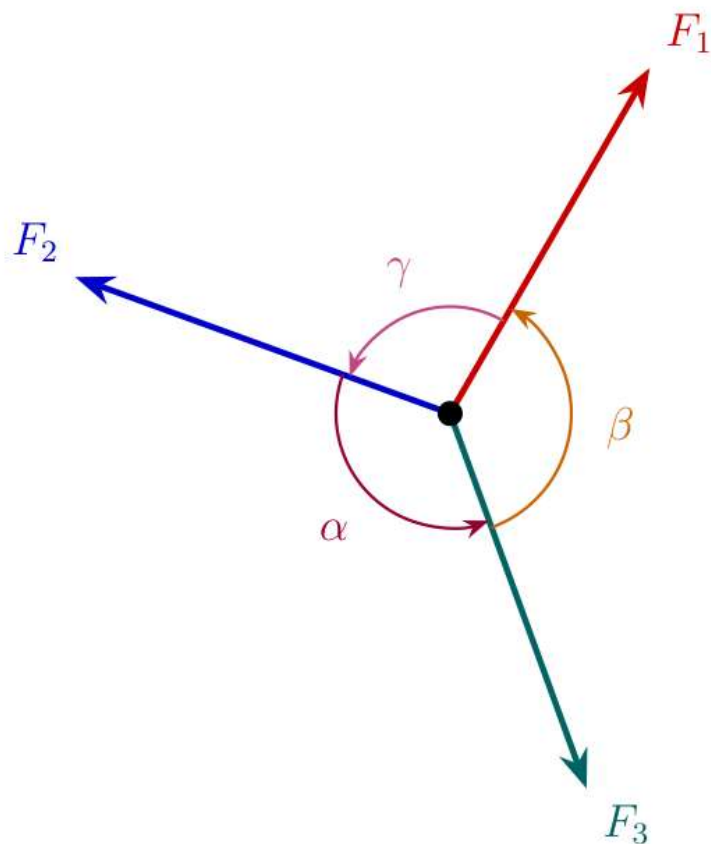
(where $g = 9.81 \text{ m/s}^2$ is the standard acceleration due to gravity).

- Tension Force (T):** A pulling force transmitted axially through a flexible cord, rope, or cable. Tension always pulls **away** from the point or joint being analyzed.
- Concurrent Forces:** A set of forces whose lines of action pass through a single common point.
- Coplanar Forces:** Forces that lie in the same two-dimensional plane.
- Static Equilibrium:** A state in which a system is at rest because the vector sum of all forces acting on it is zero ($\sum \vec{F} = 0$).

Lami's Theorem

Mathematical Statement

Lami's Theorem states that if **three coplanar, concurrent forces** acting on a body keep it in static equilibrium, then the magnitude of each force is directly proportional to the sine of the angle between the remaining two forces.



Visual Description: Three forces F_1 , F_2 , and F_3 act outward from a central origin point. The angle directly opposite to force F_1 (between F_2 and F_3) is labeled α . The angle directly opposite to force F_2 (between F_1 and F_3) is labeled β . The angle directly opposite to force F_3 (between F_1 and F_2) is labeled γ .

Mathematical Equation

$$\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$$

Where:

- F_1, F_2, F_3 are the magnitudes of the three forces.
- α is the angle between $\vec{F}_2$ and $\vec{F}_3$.
- β is the angle between $\vec{F}_1$ and $\vec{F}_3$.
- γ is the angle between $\vec{F}_1$ and $\vec{F}_2$.
- $\alpha + \beta + \gamma = 360^\circ$.

Prerequisites & Rules for Applying Lami's Theorem

To apply Lami's Theorem correctly without mathematical errors, three strict criteria must be satisfied:

1. **Exactly Three Forces:** The system at the joint or point must consist of precisely **three** forces. (If there are 4 or more forces, standard component resolution $\Sigma F_x = 0, \Sigma F_y = 0$ must be used instead).
2. **Coplanar and Concurrent:** All three force vectors must lie in the same plane and intersect at the same central point.
3. **Directional Consistency (All-Outward or All-Inward):** All three force vectors must either point **away** from the joint (divergent/pulling) or all point **towards** the joint (convergent/pushing). They cannot be mixed.

Step-by-Step Problem-Solving Methodology

When solving multi-cord or interconnected cable systems:

- 1. Identify Free Body Diagram (FBD) Nodes:** Identify concurrent points where unknown forces meet known loads or other cables.
 - 2. Convert Mass to Weight:** Convert any load given in kg to Weight in Newtons ($W = m \times 9.81$).
 - 3. Determine Geometric Angles:** Use basic geometry, trigonometry ($\tan^{-1}$, $\sin$, $\cos$), and alternate interior/complementary angle rules to find the angles between vectors.
 - 4. Isolate Concurrent Joints (Sequential Analysis):**
 - Begin at a joint with **only two unknowns** and one known force.
 - Draw the FBD for the joint.
 - Calculate internal angles opposite to each force vector.
 - Apply Lami's Theorem ratio equations to determine unknown tensions.
 - 5. Propagate Forces:** Carry calculated tension forces forward as known inputs to adjacent joints until all cord tensions are resolved.
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Comprehensive Worked Example: Five-Cord Suspended System

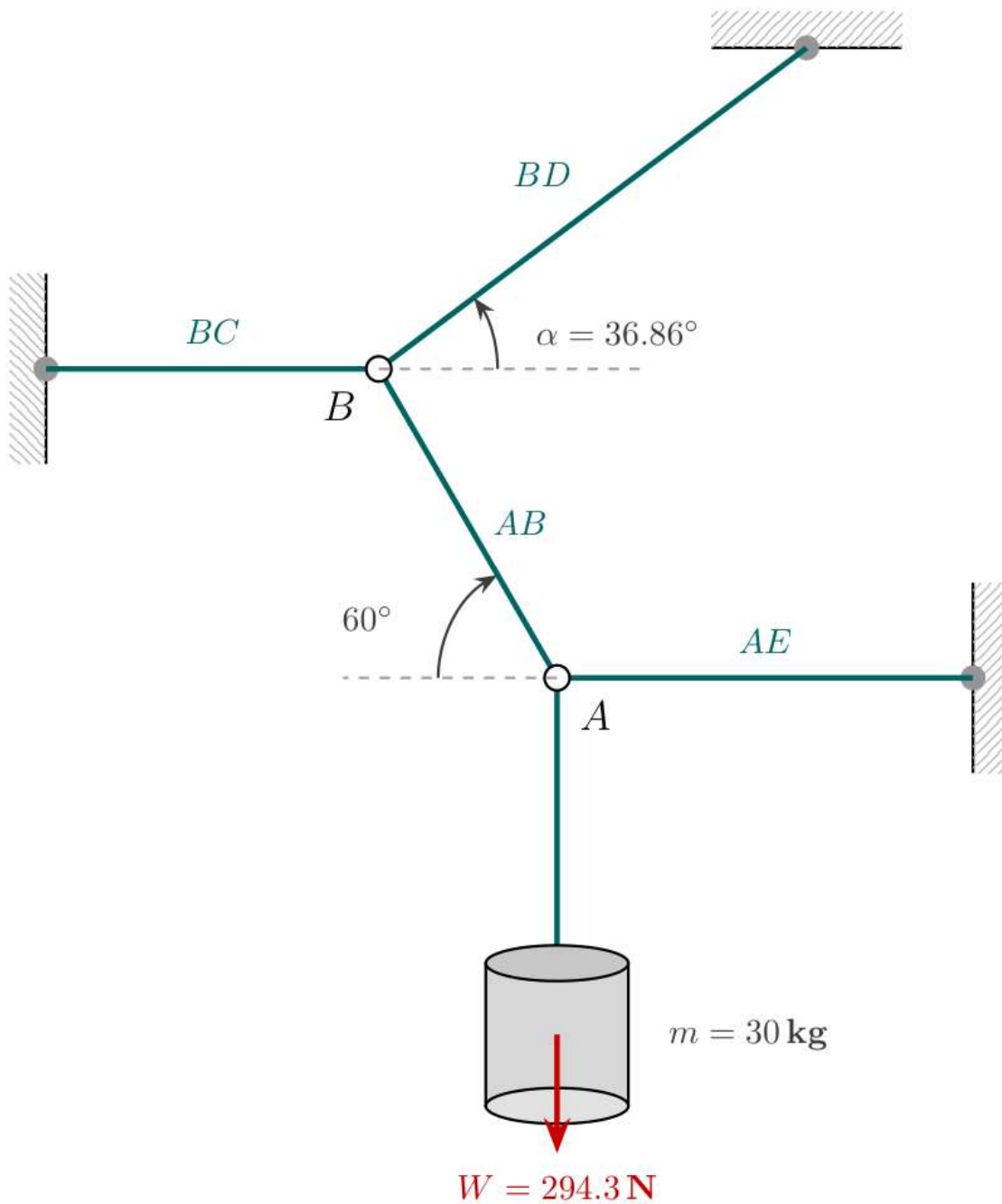
Problem Statement

A pipe with a mass of 30 kg is supported at joint A by a system of five cords (AB , AE , BC , BD , and the vertical hanging cord holding the pipe).

The geometrical configuration is given as follows:

- Joint A supports the 30 kg pipe hanging vertically downwards (W).
- Cord AE extends horizontally to the right from joint A .
- Cord AB extends upwards and to the left from joint A , inclined at an angle of 60° relative to the horizontal line at A .
- Cord BC extends horizontally to the left from joint B .
- Cord BD extends upwards and to the right from joint B , at an angle $\alpha = \tan^{-1}\left(\frac{3}{4}\right) = 36.86^\circ$ relative to the horizontal.
- Cord AB connects joint A to joint B .

Determine the tension force in each of the five cords required to maintain static equilibrium.



Visual Description: Complete structural layout showing joint A and joint B . A 30 kg pipe hangs vertically from joint A . Cord AE runs horizontally rightward to a fixed wall. Cord AB connects joint A to joint B slanting upwards and leftwards at 60° to the horizontal. From joint B , cord BC runs horizontally leftward to a wall, and cord BD extends upwards and rightwards at an angle $\alpha = 36.86^\circ$ to a ceiling support.

Phase 1: Convert Mass to Weight

Given mass $m = 30 \text{ kg}$:

$$W = m \cdot g = 30 \text{ kg} \times 9.81 \text{ m/s}^2 = 294.3 \text{ N}$$

The vertical cord directly supporting the pipe carries a tension equal to the weight of the pipe:

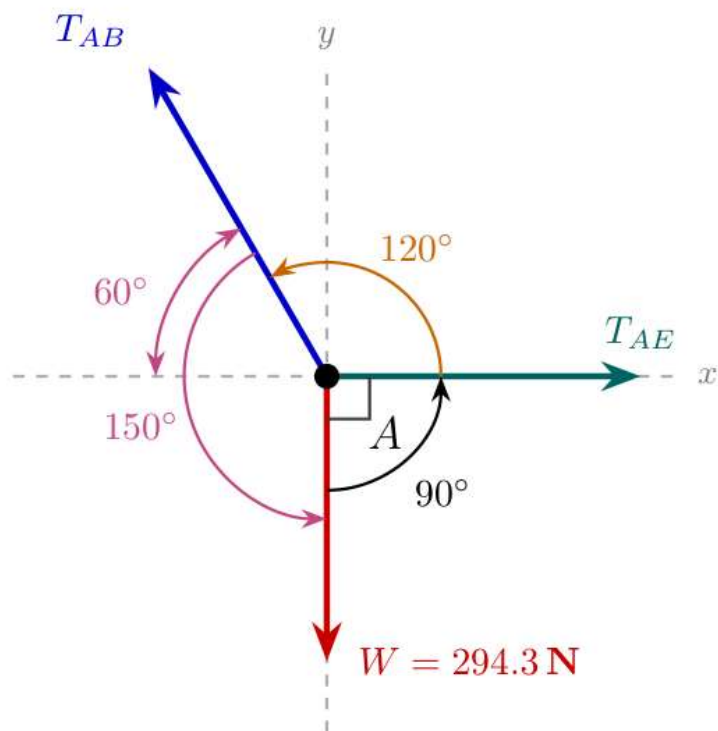
$$\mathbf{T_{\text{pipe}} = 294.3 \text{ N} \quad (\downarrow)}$$

Phase 2: Analysis of Joint A

Step 1: Draw Free Body Diagram at Joint A

At Joint A , three forces interact:

1. $W = 294.3 \text{ N}$ directed vertically downwards.
2. T_{AE} directed horizontally to the right along line AE .
3. T_{AB} directed pulling away along cord AB (upwards and to the left at 60° above the negative horizontal axis).



Visual Description: Concurrent force system at origin A . Force $W = 294.3 \text{ N}$ points down along the negative y -axis. Force T_{AE} points right along the positive x -axis. Force T_{AB} points into the second quadrant at 60° above the negative x -axis. The angle between T_{AE} and W is 90° . The angle between T_{AB} and T_{AE} is 120° . The angle between T_{AB} and W is 150° .

Step 2: Calculate Angles Between Forces at Joint A

- Angle between T_{AE} and W :

$$\theta_{AE-W} = 90^\circ$$

- Angle between T_{AB} and T_{AE} :

$$\theta_{AB-AE} = 180^\circ - 60^\circ = 120^\circ$$

- Angle between T_{AB} and W :

$$\theta_{AB-W} = 60^\circ + 90^\circ = 150^\circ$$

(Check angle sum: $90^\circ + 120^\circ + 150^\circ = 360^\circ$)

Step 3: Apply Lami's Theorem at Joint A

$$\frac{W}{\sin(\theta_{AB-AE})} = \frac{T_{AB}}{\sin(\theta_{AE-W})} = \frac{T_{AE}}{\sin(\theta_{AB-W})}$$

Substitute the known values:

$$\frac{294.3}{\sin(120^\circ)} = \frac{T_{AB}}{\sin(90^\circ)} = \frac{T_{AE}}{\sin(150^\circ)}$$

1. Solve for T_{AB} :

$$\frac{294.3}{\sin(120^\circ)} = \frac{T_{AB}}{1}$$

$$T_{AB} = \frac{294.3}{0.8660}$$

$$\mathbf{T_{AB} = 339.83 \text{ N}}$$

2. Solve for T_{AE} :

$$\frac{294.3}{\sin(120^\circ)} = \frac{T_{AE}}{\sin(150^\circ)}$$

$$T_{AE} = \frac{294.3 \times \sin(150^\circ)}{\sin(120^\circ)} = \frac{294.3 \times 0.5}{0.8660}$$

$$\mathbf{T_{AE} = 169.91 \text{ N}}$$

Phase 3: Analysis of Joint B

Step 1: Geometry & Transferred Forces

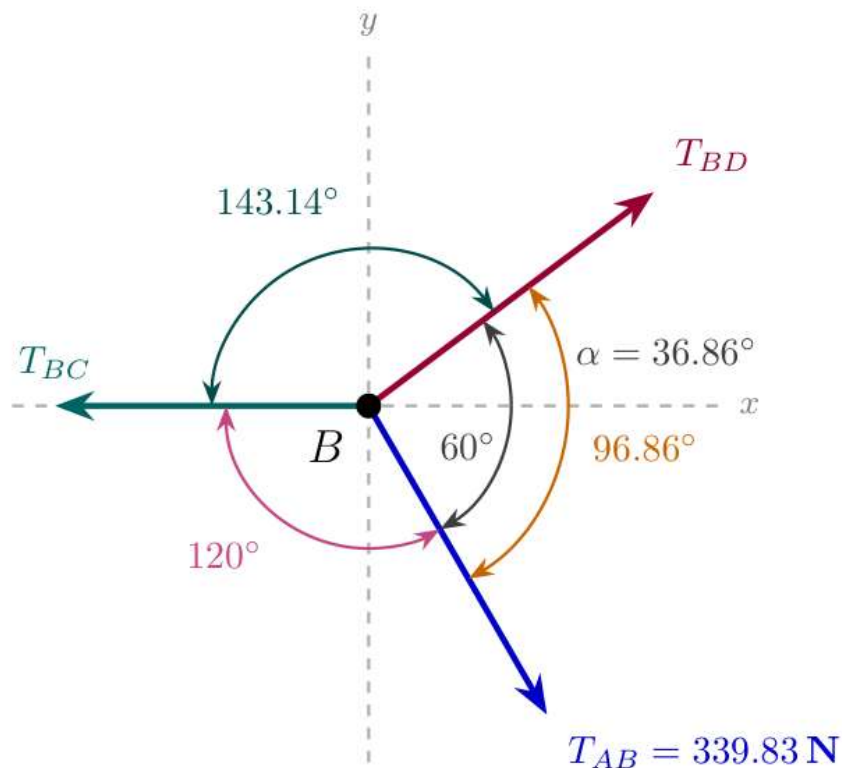
From equal and opposite action-reaction principles, the tension force in cord AB acts pulling **away** from joint B (downwards and to the right at 60° to the horizontal).

- Magnitude of transferred force: $T_{AB} = 339.83 \text{ N}$.
- Angle of cord BD : $\alpha = \tan^{-1}\left(\frac{3}{4}\right) = 36.86^\circ$ above the horizontal.

Step 2: Draw Free Body Diagram at Joint B

At Joint B , three forces interact:

1. T_{BC} directed horizontally to the left.
2. T_{BD} directed upwards and to the right at 36.86° to the horizontal.
3. $T_{AB} = 339.83 \text{ N}$ directed downwards and to the right at 60° to the horizontal.



Visual Description: Concurrent force system at origin B . Force T_{BC} points along the negative horizontal axis. Force T_{BD} points into the first quadrant at 36.86° above the positive horizontal axis. Force T_{AB} points into the fourth quadrant at 60° below the positive horizontal axis. The angle between T_{BD} and T_{AB} is 96.86° . The angle between T_{BC} and T_{AB} is 120° . The angle between T_{BC} and T_{BD} is 143.14° .

Step 3: Calculate Angles Between Forces at Joint B

- Angle between T_{BD} and T_{AB} (γ):

$$\gamma = 36.86^\circ + 60^\circ = 96.86^\circ$$

- Angle between T_{BC} and T_{AB} (β):

$$\beta = 180^\circ - 60^\circ = 120^\circ$$

- Angle between T_{BC} and T_{BD} (α):

$$\alpha = 180^\circ - 36.86^\circ = 143.14^\circ$$

(Check angle sum: $96.86^\circ + 120^\circ + 143.14^\circ = 360^\circ$)

Step 4: Apply Lami's Theorem at Joint B

$$\frac{T_{AB}}{\sin(\theta_{BC-BD})} = \frac{T_{BD}}{\sin(\theta_{BC-AB})} = \frac{T_{BC}}{\sin(\theta_{BD-AB})}$$

$$\frac{339.83}{\sin(143.14^\circ)} = \frac{T_{BD}}{\sin(120^\circ)} = \frac{T_{BC}}{\sin(96.86^\circ)}$$

1. Solve for T_{BD} :

$$T_{BD} = \frac{339.83 \times \sin(120^\circ)}{\sin(143.14^\circ)}$$

$$\sin(143.14^\circ) \approx 0.5999, \quad \sin(120^\circ) \approx 0.8660$$

$$T_{BD} = \frac{339.83 \times 0.8660}{0.5999}$$

$$\mathbf{T_{BD} = 490.58\,N}$$

2. Solve for T_{BC} :

$$T_{BC} = \frac{339.83 \times \sin(96.86^\circ)}{\sin(143.14^\circ)}$$

$$\sin(96.86^\circ) \approx 0.9928$$

$$T_{BC} = \frac{339.83 \times 0.9928}{0.5999}$$

$$\mathbf{T_{BC} = 562.39\,N}$$

Final Results Summary Table

Cord Designation	Structural Role / Connection	Force Magnitude (N)	Direction / Orientation
Vertical Load Cord	Suspends 30 kg pipe directly	294.30 N	Vertically Downward (270°)
Cord AE	Connects Joint A to Wall E	169.91 N	Horizontally Rightward (0°)
Cord AB	Connects Joint A to Joint B	339.83 N	Inclined 60° to horizontal
Cord BD	Connects Joint B to Ceiling D	490.58 N	Inclined 36.86° to horizontal
Cord BC	Connects Joint B to Wall C	562.39 N	Horizontally Leftward (180°)